

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07

COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

D. Sindhu.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is

lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis

Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Answers:

1. File Transfer (150 MB)

Given:

- File size = 150 MB = $150 \times 1024 \times 1024 = 157,286,400$ bytes
- Segment size = 1500 bytes
- Data Link overhead/frame = 40 bytes
- Bandwidth = 50 Mbps

Step 1: Number of segments

$$\frac{157,286,400}{1500} = 104,857.6$$

Round up:

Segments = 104,858

Each segment becomes one frame.

Frames = 104,858

Step 2: Total Data Link overhead

$$104,858 \times 40 = 4,194,320 \text{ bytes}$$

≈ 4 MB

Step 3: Total transmission size

$$157,286,400 + 4,194,320 = 161,480,720 \text{ bytes}$$

Convert to bits:

$$161,480,720 \times 8 = 1,291,845,760 \text{ bits}$$

Transmission time:

$$\frac{1,291,845,760}{50 \times 10^6} = 25.84 \text{ s}$$

Answer

- Segments = 104,858
- Frames = 104,858
- Overhead = 4,194,320 bytes

- Transmission time ≈ 25.84 s

Explanation

- Transport Layer: Segmentation, sequencing, acknowledgments, retransmission.
- Data Link Layer: Framing, MAC addressing, error detection (CRC).

2. Sliding Window

Given

- Window size = 6
- Total frames = 48

Windows required

$$48 \div 6 = 8$$

Windows = 8

Retransmissions

One frame lost in every window.

$$8 \times 1 = 8$$

Retransmissions = 8

Explanation

Flow control prevents the sender from overwhelming the receiver, improving network efficiency and reducing congestion.

3. IP Fragmentation

Given

- IP packet = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Maximum data per fragment:

$$1500 - 20 = 1480 \text{ bytes}$$

Number of fragments

$$\frac{12000}{1480} = 8.11$$

Round up:

Fragments = 9

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = 9
- Header overhead = 180 bytes

Explanation

The Network Layer fragments packets, assigns fragment offsets and identification numbers, and reassembles them at the destination.

4. Sliding Window

Given

- Window size = 6
- Total frames = 48

Windows required

$$48 \div 6 = 8$$

Windows = 8

Retransmissions

One frame lost in every window.

$$8 \times 1 = 8$$

Retransmissions = 8

Answer

- Windows = 8
- Retransmissions = 8

Explanation

Flow control ensures efficient communication by matching sender speed with receiver capacity.

5. Session Layer Synchronization

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints before failure

$$390 \div 60 = 6.5$$

Completed checkpoints:

6 checkpoints

Last checkpoint = 360 MB

Retransmission

$$390 - 360 = 30 \text{ MB}$$

Answer

- Checkpoints = 6
- Retransmitted data = 30 MB

Explanation

Synchronization checkpoints allow communication to resume from the last checkpoint instead of restarting the entire transfer.

6. Presentation Layer

Given

- Original file = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Percentage reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed size = 540 MB
- Encrypted size = 567 MB
- Total reduction = 37%

Explanation

The Presentation Layer compresses data to reduce transmission time and encrypts it to provide confidentiality.

7. FTP Transfer

Given

- Data = 3 GB = 3072 MB
- Request size = 3 MB
- Throughput = 120 Mbps

Requests

$$3072 \div 3 = 1024$$

Requests = 1024

Transmission time

3072 MB =

$$\frac{3072 \times 8}{120} = 204.8 \text{ s}$$

Answer

- Requests = 1024
- Transmission time $\approx 204.8 \text{ s}$

Explanation

The Application Layer provides services such as FTP, authentication, file management, and error reporting.

8. End-to-End Delay

Processing delays:

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Total processing:

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Propagation delay = 18 ms

Transmission delay = 12 ms

Total delay:

$$15 + 18 + 12 = 45 \text{ ms}$$

Answer

- Total end-to-end delay = 45 ms

Explanation

- Application: User services
- Transport: Reliability
- Network: Routing
- Data Link: Framing
- Physical: Signal transmission

9. Frame Overhead

Given

- Useful data = 80 MB = 83,886,080 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes

Useful data/frame:

$$1600 - 80 = 1520 \text{ bytes}$$

Frames

$$\frac{83,886,080}{1520} = 55188.2$$

Round up:

Frames = 55,189

Total overhead

$$55189 \times 80 = 4,415,120 \text{ bytes}$$

Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = 55,189
- Overhead = 4,415,120 bytes
- Efficiency = 95%

Explanation

Protocol overhead consumes bandwidth, reducing effective data throughput.

10. Hospital IoT Data Transfer

Given

- Original file = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link overhead = 60 bytes/frame

Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

Compressed bytes:

$$168 \times 1024 \times 1024 = 176,160,768 \text{ bytes}$$

Segments

$$\frac{176,160,768}{1400} = 125829.12$$

Round up:

Segments = 125,830

Protocol overhead

$$125830 \times 60 = 7,549,800 \text{ bytes}$$

Final transmitted data

$$176,160,768 + 7,549,800 = 183,710,568 \text{ bytes}$$

$\approx 175.2 \text{ MB}$

Answer

- Compressed size = 168 MB
- Segments = 125,830
- Protocol overhead = 7,549,800 bytes
- Final transmitted data = 183,710,568 bytes ($\approx 175.2 \text{ MB}$)

Explanation

- Presentation Layer: Compression and encryption.
- Transport Layer: Segmentation, sequencing, reliability.
- Data Link Layer: Framing and error detection.
- Physical Layer: Transmits bits over the communication medium.